

Hongcheng Han (韩泓丞)

✉ hanhc@stu.xjtu.edu.cn ✉ @Mainsail9999
🌐 <http://hongchenghan.github.io>
🏠 No.28 Xianning West Road, Xi'an 710049, Shaanxi, China



Biography

Hongcheng Han is a Ph.D. candidate in Artificial Intelligence, specializing in medical image analysis and multi-modal medical data fusion. Holding a Bachelor's degree in Energy and Power Engineering with a background in mechanical design, he brings a strong interdisciplinary foundation to his research. He has also contributed to automotive intelligent cockpit projects, designing algorithms for driver behavior and state analysis.

Education

- 09/2022 – present **Ph.D. Candidate, Control Science and Engineering**
Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University, Xi'an, China
- 03/2026 – 03/2027 **Visiting PhD Student (Funded)**
College of Design and Engineering, National University of Singapore, Singapore
- 09/2020 – 06/2022 **M.Eng., Software Engineering (Transfer to Ph.D. in 2022.09)**
School of Software Engineering, Xi'an Jiaotong University, Xi'an, China
- 09/2016 – 06/2020 **B.Eng., Energy and Power Engineering**
College of Energy and Power Engineering, Xi'an Jiaotong University, Xi'an, China

Research Publications

Journal Articles

- 1 **H. Han**, S. Du, and J. Wang, "Virtual contrast enhanced ultrasound synthesis for prostate cancer diagnosis," *IEEE Transactions on Medical Imaging*, 2026, (Under review).
- 2 **H. Han**, Z. Tian, M. Wang, *et al.*, "AsyCMST: Asymmetric cross-modal spatio-temporal learning for multimodal ultrasound nodule recognition," *Medical Image Analysis*, p. 104 127, 2026.
- 3 **H. Han**, Z. Tian, Q. Guo, J. Jiang, S. Du, and J. Wang, "HSC-T: B-ultrasound-to-elastography translation via hierarchical structural consistency learning for thyroid cancer diagnosis," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 2, pp. 799–806, 2025.
- 4 **H. Han**, Y. Lin, Q. Guo, *et al.*, "Dermoscopic image classification for multi-modal medical corpus," *Journal of Northwest University (Natural Science Edition)*, vol. 53, no. 3, pp. 2817–2830, 2023.
- 5 X. Wang[#], **H. Han[#]**, M. Xu, *et al.*, "STNet: Shape and texture joint learning through two-stream network for knowledge-guided image recognition," *Frontiers in Neuroscience*, vol. 17, p. 1 212 049, 2023, (Co-first author).
- 6 D. Zhang[#], **H. Han[#]**, S. Du, *et al.*, "MPMR: Multi-scale feature and probability map for melanoma recognition," *Frontiers in Medicine*, vol. 8, p. 775 587, 2022, (Co-first author).

Conference Proceedings

- 1 **H. Han**, Z. Tian, Q. Guo, J. Jiang, J. Wang, and S. Du, "HSC-T: B-ultrasound-to-elastography translation via hierarchical structural consistency learning for thyroid cancer diagnosis," in *IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)*, (Oral presentation, one of the top-rate papers), IEEE, 2024. 🌐 URL: <https://openreview.net/forum?id=EwSohBIcje>.

- 2 **H. Han**, Z. Tian, Y. Liu, S. Li, D. Zhang, and S. Du, "Gsa-gaze: Generative self-adversarial learning for domain generalized driver gaze estimation," in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, (Oral presentation), IEEE, 2023, pp. 1610–1615.
- 3 **H. Han**, S. Du, D. Zhang, H. Long, and Y. Guo, "Precise dental staging method through panoramic radiographs based on deep learning," in *2020 Chinese Automation Congress (CAC)*, IEEE, 2020, pp. 377–386.

Skills

Languages	■ Good reading, writing and speaking competencies for English and Mandarin Chinese.
Research	■ Machine learning algorithm design, digital image processing.
Coding	■ Python (Pytorch), C++.
Misc.	■ $\text{\LaTeX}$, graphic design (AI, PS, LR), video editing (PR), 3D modeling (Solidworks).

Awards and Achievements

- 2025 ■ **3rd Prize**, China Post-Graduate Mathematical Contest in Modeling.

Awards in Science and Technology

- 2024 ■ **1st Prize**, Asia and Pacific Mathematical Contest in Modeling (APMCM).
 ■ **3rd Prize**, China Post-Graduate Mathematical Contest in Modeling.
- 2021 ■ **2nd Prize**, Asia and Pacific Mathematical Contest in Modeling (APMCM).
- 2019 ■ **Overall Championship**, SAE International College Aircraft Design Competition (Micro Class), Los Angeles, California, USA.
- 2017 ■ **1st Prize**, China International Aircraft Design Challenge (CADC). Zhenjiang, China.

Awards in Innovation and Entrepreneurship

- 2024 ■ **2nd Prize**, 19th China Graduate Student Electronic Design Competition Business Plan Special Competition.
- 2021 ■ **1st Prize**, 16th China Graduate Student Electronic Design Competition Business Plan Special Competition.
 ■ **3rd Prize**, 9th XJTU Entrepreneurship Practice Competition, Xi'an, China.

Scholastic Achievements

- 2020–2025 ■ **Top-class Academic Scholarship**, XJTU.
- 2025 ■ **Outstanding Student Award**, XJTU.
 ■ **Weichai Power Scholarship**, XJTU.
- 2024 ■ **Outstanding Student Award**, XJTU.
- 2021 ■ **Outstanding Student Award**, XJTU.